

Supplementary Material for: PCSC: Pareto-Coverage Stress Coevolution for Robust Portfolio Optimization.

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1 Data Audit and Preprocessing

1.1 Data Source and Price Adjustment

The data are obtained from BaoStock and comprise daily OHLCV observations for China A-share stocks from 2016–2025. The download script uses `adjustflag=2`; therefore, prices are forward-adjusted, while volume is the number of shares traded as returned by the interface. Repeated queries with identical parameters show that price levels may be proportionally rescaled across query dates, whereas daily returns remain identical within numerical tolerance. Accordingly, supplementary history is merged with the original data only in return space, and forward-adjusted price levels are not used for cross-sectional comparisons across stocks.

1.2 Point-in-Time Security List and Survivorship-Bias Control

The original data were downloaded using the security list available at the end of 2023 and may therefore omit stocks that had previously been delisted or ceased trading. To reduce survivorship bias, we supplement the missing history using the point-in-time security list as of the end of 2018; all stock-universe eligibility criteria and ranking variables are restricted to information available by December 31, 2018. Candidate stocks must contain at least 500 historical records, with trading-day record coverage of at least 95% and positive-volume-day coverage of at least 90% in 2018. Among eligible stocks, a fixed universe of 50 stocks is selected in descending order of median volume on positive-volume days in 2018, with ticker codes used for deterministic tie-breaking. This measure serves only as a liquidity proxy and does not represent actual order-book liquidity. Survival status over 2019–2025 is not used to screen stocks.

Table 1 reports the frozen stock universe and its ranking order. Repairing the point-in-time list adds `sz_300104` to the universe; this stock is unavailable for trading on most month-end decision dates, while `sh_601989` has no new quotes after August 2025. Neither stock is removed retrospectively based on its survival status during the test period; instead, both are handled using the availability mask described in the next subsection.

1.3 Structural Quality and Alignment Rules

The full structural audit checks field parsing, duplicate dates, temporal ordering, missing values, price validity, and OHLC consistency. Zero-volume records are retained and used as a proxy for suspension or unavailability for trading. Stocks are aligned to the market trading calendar. Missing closing prices are forward-filled solely to maintain valuation continuity, while the original-quote and positive-volume masks are retained. Filled values are not treated as tradable prices, and pre-listing observations are not backfilled. If a stock has no valid quote or positive volume on a decision day, its weight cap is set to zero for that optimization. If no new valid quote has appeared since the previous monthly execution, the stock is marked unavailable at the current decision, followed by a modeled exit and reallocation at its last available closing price. This rule uses only the missingness status known at the decision time, but an exit at the last price should not be interpreted as an actual delisting transaction. Affected stocks and dates are reported separately in the data manifest.

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Table 1: Frozen 50-stock universe as of the end of 2018, ordered by median volume on positive-volume days in descending order

1–10	11–20	21–30	31–40	41–50
sz_000725	sh_600050	sh_600031	sz_000425	sh_601166
sh_601288	sh_601939	sz_002195	sh_600808	sz_002340
sh_601398	sz_000001	sh_601318	sz_000932	sh_600567
sz_300104	sh_601668	sh_601328	sh_600887	sz_000717
sh_601899	sh_600030	sh_601989	sh_601111	sh_600115
sz_300059	sh_600048	sz_000100	sh_600256	sz_000636
sh_601988	sh_600029	sh_600282	sz_002797	sz_000002
sh_603993	sz_000735	sh_601225	sz_000651	sh_601933
sh_600028	sz_002027	sh_600019	sz_002129	sz_000338
sh_600010	sh_601818	sh_600016	sh_600782	sz_000709

2 Complete Mathematical Definitions

This section first defines the outer market-time protocol for monthly Walk-forward Evaluation and then presents the mathematical definition of PCSC at a single decision time. The monthly rebalancing period is indexed by m , the trading day by t , and the evolutionary generation by the superscript g ; these indices represent the interperiod loop, market time, and algorithmic time, respectively.

2.1 Monthly Walk-Forward Timing Protocol

In the m th rebalancing period, let D_m denote the close of the decision day, E_m the execution time at the close of the next trading day, and S_m the first return-accrual day after execution. Let $\text{next}(\cdot)$ denote the next trading day in the market calendar. Then,

$$E_m = \text{next}(D_m), \quad S_m = \text{next}(E_m), \quad t = D_m \quad (1)$$

A single PCSC run uses only the most recent T return observations available by the close of D_m and the portfolio state known at that time. The algorithm generates and freezes the target weights after D_m and executes them at the close of E_m ; the return from D_m to E_m is not used to generate or reoptimize the current target weights.

The out-of-sample holding period for the new weights is given by the set of return dates $\mathcal{H}_m$:

$$\mathcal{H}_m = \{S_m, \text{next}(S_m), \dots, E_{m+1}\}, \quad d_t = |\mathcal{H}_m|, \quad t = D_m \quad (2)$$

Daily returns are labeled by their ending close; therefore, the close-to-close return ending on E_{m+1} is still earned by the period- m portfolio. The next target portfolio becomes effective after rebalancing at the close of E_{m+1} . If no subsequent execution day exists for the final evaluation period, the holding interval is truncated at the prespecified and frozen sample end date. This convention ensures that adjacent holding periods neither overlap nor omit investable trading days.

2.2 Return Sample and Rebalancing Decision

Suppose that the stock universe contains n stocks. Following the convention in Section 2.1, reindex the most recent T trading days available at the decision time $t = D_m$ as $\ell = 1, \dots, T$, and let $P_{\ell,i}$ denote the forward-adjusted closing price of stock i . The simple return $r_{\ell,i}$, daily return vector $\mathbf{r}_\ell$, historical return matrix $\mathbf{R}_t$, and its sample mean $\bar{\mathbf{r}}_t$ are defined as follows:

$$\begin{aligned} r_{\ell,i} &= \frac{P_{\ell,i}}{P_{\ell-1,i}} - 1 \\ \mathbf{r}_\ell &= (r_{\ell,1}, \dots, r_{\ell,n})^\top \\ \mathbf{R}_t &= (\mathbf{r}_1, \dots, \mathbf{r}_T)^\top \in \mathbb{R}^{T \times n} \\ \bar{\mathbf{r}}_t &= \frac{1}{T} \sum_{\ell=1}^T \mathbf{r}_\ell \end{aligned} \quad (3)$$

PCSC uses only the return matrix $\mathbf{R}_t$ for stress construction and objective evaluation. Volume is used only as a liquidity proxy when constructing the stock universe; it does not enter the stressed-return transformation and is not interpreted as order-book liquidity. The length d_t of the next holding period is determined by the trading calendar in Eq. (2). Apart from such prespecified backtest settings, all inputs to the current decision are obtained from time t or earlier; realized returns in the next period do not participate in the current optimization.

2.3 Portfolios, Turnover, and the Feasible Set

The j th portfolio is represented as

$$\mathbf{w}_j = (w_{j,1}, \dots, w_{j,n})^\top \quad (4)$$

Let $\mathbf{w}_t^- \equiv \mathbf{w}_m^{\text{dec},-}$ denote the drifted weights known at the close of decision day D_m . Let $\chi_{t,i} \in \{0, 1\}$ indicate the availability of asset i at the decision time, and define the time-varying weight cap $u_{t,i} = u\chi_{t,i}$, where u is the nominal per-asset cap. We use one-way turnover:

$$\tau(\mathbf{w}_j, \mathbf{w}_t^-) = \frac{1}{2} \|\mathbf{w}_j - \mathbf{w}_t^-\|_1 \quad (5)$$

Under the long-only constraint, a unit budget, time-varying per-asset weight caps $u_{t,i}$, and maximum allowable turnover $\tau_{\max}$, the feasible set is

$$\mathcal{W}_t = \{\mathbf{w} \in \mathbb{R}^n \mid \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq u_{t,i}, \quad \tau(\mathbf{w}, \mathbf{w}_t^-) \leq \tau_{\max}\} \quad (6)$$

The turnover constraint in the feasible set is an ex ante constraint computable at the decision time. An unavailable asset satisfies $u_{t,i} = 0$ and therefore cannot be held in the new target portfolio. Feasibility requires at least $\sum_i u_{t,i} \geq 1$. When $\mathbf{w}_t^-$ does not exist at initial investment, the turnover cap is not imposed; its one-way turnover is recorded as 1 for transaction-cost calculation.

Given a one-way proportional transaction cost c , the approximate expected net return of the portfolio over the holding period is

$$\hat{R}_t^{\text{net}}(\mathbf{w}_j) = d_t \bar{\mathbf{r}}_t^\top \mathbf{w}_j - c\tau(\mathbf{w}_j, \mathbf{w}_t^-) \quad (7)$$

This term is a historical-mean estimate used in the optimization objective, rather than a supervised prediction of next-period returns.

Let $\mathbf{w}_m^*$ denote the target weights selected by PCSC at D_m . The previous portfolio remains invested until the close of E_m . If $\mathbf{r}_{E_m}$ denotes the vector of asset returns from the close of D_m to the close of E_m , the drifted weights immediately before rebalancing on the execution day are

$$w_{m,i}^{\text{exe},-} = \frac{w_{t,i}^-(1 + r_{E_m,i})}{1 + (\mathbf{w}_t^-)^\top \mathbf{r}_{E_m}}, \quad i = 1, \dots, n \quad (8)$$

The realized one-way turnover and the corresponding transaction cost are, respectively,

$$\tau_m^{\text{real}} = \frac{1}{2} \|\mathbf{w}_m^* - \mathbf{w}_m^{\text{exe},-}\|_1, \quad \text{TC}_m = c\tau_m^{\text{real}} \quad (9)$$

Equations (5) and (6) use the decision-day weights, whereas Eq. (9) is used in the out-of-sample backtest. The target weights are not reoptimized using $\mathbf{r}_{E_m}$. Because prices drift for one trading day between the decision and execution dates, τ_m^{real} may differ slightly from the ex ante turnover. We report realized turnover and realized costs without characterizing the ex ante turnover cap as an absolute post-execution hard constraint.

2.4 Historical Blocks and Stress Scenarios

The b th unperturbed historical block of length H is defined in Eq. 10, and the set of all unperturbed historical blocks is given in Eq. 11:

$$\mathbf{B}_{t,b} = \begin{bmatrix} \mathbf{r}_{b+1}^\top \\ \vdots \\ \mathbf{r}_{b+H}^\top \end{bmatrix}, \quad b \in \mathcal{I}_t = \{0, \dots, T - H\} \quad (10)$$

$$\mathcal{B}_t = \{\mathbf{B}_{t,b} : b \in \mathcal{I}_t\}, \quad B = |\mathcal{B}_t| = T - H + 1 \quad (11)$$

The set $\mathcal{B}_t$ is fixed at the current rebalancing time and does not evolve. It is used primarily to construct a separate historical loss baseline for each portfolio. The stress-path length H controls the time scale of historical risk evaluation, whereas d_t is the out-of-sample holding-period length determined by the market trading calendar. They serve different purposes and need not be equal.

A stress scenario $s_k = (b_k, a_k, \rho_k)$ consists of a historical block starting index b_k , a multiplier a_k for the common market component, and a residual shrinkage coefficient ρ_k . Its feasible space is given in Eq. 12, where $0 < a_{\min} \leq a_{\max}$ and $0 \leq \rho_{\min} \leq \rho_{\max} \leq 1$.

$$\Omega_t = \{(b, a, \rho) \mid b \in \mathcal{I}_t, \quad a_{\min} \leq a \leq a_{\max}, \quad \rho_{\min} \leq \rho \leq \rho_{\max}\} \quad (12)$$

Within historical block $\mathbf{B}_{t,b}$, the equally weighted common market component on day h and the residual of stock i are, respectively,

$$m_{b,h} = \frac{1}{n} \sum_{i=1}^n r_{b+h,i} \quad e_{i,b,h} = r_{b+h,i} - m_{b,h} \quad (13)$$

Given scenario s_k , the stressed return of stock i on scenario day h is

$$r_{i,h}^{(s_k)} = a_k m_{b_k,h} + (1 - \rho_k) e_{i,b_k,h}, \quad h = 1, \dots, H \quad (14)$$

a_k changes the magnitude of the common market component, while ρ_k shrinks stock-specific residuals toward that component. In particular, $(b, 1, 0)$ recovers the unperturbed historical block $\mathbf{B}_{t,b}$. This representation preserves the temporal structure of actual historical paths while allowing only two bounded continuous perturbations. Randomly generated candidates are not necessarily all adverse scenarios; their stress intensity is determined by the subsequent loss and coverage evaluations.

2.5 Path Loss and Portfolio Optimization Objectives

The day- h return of portfolio $\mathbf{w}_j$ under stress scenario s_k and the additive cumulative return from the beginning of the path through day q are, respectively,

$$x_{j,h}^{(s_k)} = \mathbf{w}_j^\top \mathbf{r}_h^{(s_k)}, \quad z_{j,q}^{(s_k)} = \sum_{h=1}^q x_{j,h}^{(s_k)}, \quad z_{j,0}^{(s_k)} = 0 \quad (15)$$

The path loss is defined as the maximum peak-to-trough cumulative loss:

$$\begin{aligned} L(\mathbf{w}_j, s_k) &= \max \left\{ 0, \max_{0 \leq p < q \leq H} \left(z_{j,p}^{(s_k)} - z_{j,q}^{(s_k)} \right) \right\} \\ &= \max \left\{ 0, \max_{0 \leq p < q \leq H} \left[- \sum_{h=p+1}^q \mathbf{w}_j^\top \mathbf{r}_h^{(s_k)} \right] \right\} \end{aligned} \quad (16)$$

Equation (16) is a maximum path-drawdown proxy based on additive simple returns. It is used to compare scenarios during optimization and is not equivalent to the Maximum Drawdown metric computed from compounded portfolio values in the backtest.

Given a nonempty scenario archive $\mathcal{A}^g = (s_k)_{k \in \mathcal{K}^g}$ of capacity M at generation g , the two portfolio minimization objectives are

$$f_1(\mathbf{w}_j) = -d_t \bar{\mathbf{r}}_t^\top \mathbf{w}_j + c\tau(\mathbf{w}_j, \mathbf{w}_t^-), \quad f_2(\mathbf{w}_j; \mathcal{A}^g) = \max_{s_k \in \mathcal{A}^g} L(\mathbf{w}_j, s_k) \quad (17)$$

The first objective is equivalent to maximizing expected return net of transaction costs, while the second minimizes the worst path loss under the current archive. PCSC does not introduce a third predictive or risk objective. The objective function is therefore written as

$$\min_{\mathbf{w}_j \in \mathcal{W}_t} \mathbf{F}(\mathbf{w}_j; \mathcal{A}^g) = (f_1(\mathbf{w}_j), f_2(\mathbf{w}_j; \mathcal{A}^g)) \quad (18)$$

2.6 Portfolio Population and Pareto Set

The portfolio population at generation g is denoted by

$$\mathcal{P}^g = \{\mathbf{w}_j^g\}_{j=1}^N \subseteq \mathcal{W}_t \quad (19)$$

Given scenario archive $\mathcal{A}^g$, suppose that two portfolios $\mathbf{w}_a, \mathbf{w}_b \in \mathcal{P}^g$ satisfy

$$\begin{aligned} f_1(\mathbf{w}_a) &\leq f_1(\mathbf{w}_b) \\ f_2(\mathbf{w}_a; \mathcal{A}^g) &\leq f_2(\mathbf{w}_b; \mathcal{A}^g) \end{aligned} \quad (20)$$

with at least one of the two inequalities holding strictly. Then $\mathbf{w}_a$ is said to dominate $\mathbf{w}_b$.

The current Pareto set is defined as

$$\mathcal{P}_g^* = \{\mathbf{w} \in \mathcal{P}^g : \nexists \mathbf{w}' \in \mathcal{P}^g \text{ such that } \mathbf{w}' \text{ dominates } \mathbf{w}\} \quad (21)$$

The Pareto set here lies in the decision space; its objective vectors constitute the Pareto front in the objective space. Because f_2 depends on $\mathcal{A}^g$, changes to the archive may alter both the dominance relation and the Pareto set.

2.7 Historical Baseline and Portfolio-wise Normalized Vulnerability

For a portfolio $\mathbf{w}_j$ to be evaluated, its loss on the b th unperturbed historical block is denoted by

$$\ell_{j,b} = L(\mathbf{w}_j, (b, 1, 0)), \quad b \in \mathcal{I}_t \quad (22)$$

PCSC separately computes the median loss and unscaled median absolute deviation for each portfolio:

$$\begin{aligned} \mu_j &= \text{median}_{b \in \mathcal{I}_t} \ell_{j,b} \\ \sigma_j &= \text{median}_{b \in \mathcal{I}_t} |\ell_{j,b} - \mu_j| \end{aligned} \quad (23)$$

The Portfolio-wise Normalized Vulnerability of portfolio $\mathbf{w}_j$ to candidate scenario s_k is defined as

$$v_+(\mathbf{w}_j, s_k) = \max \left\{ 0, \frac{L(\mathbf{w}_j, s_k) - \mu_j}{\sigma_j + \epsilon} \right\}, \quad \epsilon > 0 \quad (24)$$

This quantity compares the candidate stress loss with the historical loss scale of the same portfolio. Truncation at zero means that a candidate below the historical median receives no positive vulnerability; ϵ handles cases in which $\sigma_j = 0$ or is extremely small. This portfolio-wise normalization is intended to reduce the influence of differences in absolute risk scales across portfolios on scenario evaluation.

2.8 Candidate Scenarios, Scenario Archive, and Vulnerability Coverage

During an archive update, the temporary candidate scenarios are represented as an ordered sequence:

$$\mathcal{S}^g = (s_1, \dots, s_Q), \quad s_k \in \Omega_t \quad (25)$$

Select M candidate indices to form $\mathcal{K}^g$. The corresponding scenarios constitute the formal archive $\mathcal{A}^g$ in Eq. 26, where $|\mathcal{K}^g| = M$ and $M \leq Q$:

$$\mathcal{A}^g = (s_k)_{k \in \mathcal{K}^g} \quad (26)$$

$\mathcal{S}^g$ exists only during the current update, whereas $\mathcal{A}^g$ is used to evaluate portfolio objectives until the next archive update. The set of unperturbed historical blocks $\mathcal{B}_t$ is used only in Eq. (23) and should not be confused with either of these objects.

Let $\mathcal{U}^g = \{\mathbf{w}_j\}_{j=1}^J$ denote the portfolio feedback set for the current archive update. At initialization, $\mathcal{U}^0 = \mathcal{P}^0$; thereafter, the current first non-dominated front is used. The Vulnerability Coverage of an arbitrary scenario set $\mathcal{A}$ over the feedback set is

$$C(\mathcal{A}; \mathcal{U}^g) = \frac{1}{J} \sum_{j=1}^J \max_{s_k \in \mathcal{A}} v_+(\mathbf{w}_j, s_k), \quad C(\emptyset; \mathcal{U}^g) = 0 \quad (27)$$

The marginal coverage contribution of a candidate scenario $s_k \notin \mathcal{A}$ relative to the current archive is

$$\Delta(s_k | \mathcal{A}, \mathcal{U}^g) = C(\mathcal{A} \cup \{s_k\}; \mathcal{U}^g) - C(\mathcal{A}; \mathcal{U}^g) \quad (28)$$

The maximum in the coverage function is computed separately for each portfolio. Thus, even if two scenarios both induce large losses, the second scenario may have a small marginal contribution if both primarily expose the same weaknesses in the same portfolios. PCSC encourages functional complementarity in vulnerability coverage, rather than geometric distance between scenario parameters.

2.9 Final Portfolio for Execution

After the algorithm terminates, suppose that the final Pareto set contains J_G portfolios. For each of the two minimization objectives, compute

$$\Delta_\ell^f = \max_{1 \leq j \leq J_G} f_{\ell j} - \min_{1 \leq j \leq J_G} f_{\ell j}, \quad \ell = 1, 2 \quad (29)$$

and define

$$\tilde{f}_{\ell j} = \begin{cases} \frac{f_{\ell j} - \min_q f_{\ell q}}{\Delta_\ell^f}, & \Delta_\ell^f > 0 \\ 0, & \Delta_\ell^f = 0 \end{cases} \quad (30)$$

The portfolio selected for execution is the Pareto portfolio j^* closest to the normalized ideal point $(0, 0)$:

$$j^* = \arg \min_{1 \leq j \leq J_G} \sqrt{\tilde{f}_{1j}^2 + \tilde{f}_{2j}^2} \quad (31)$$

If distances are equal, the portfolio with the smaller index is selected to ensure determinism. This rule only selects an executable solution from the final Pareto set and does not participate in the evolution of the scenario archive.

3 Complete PCSC Algorithm and Constraint Repair

This section describes PCSC in the order in which it is executed during a monthly rebalancing period. The algorithm uses the portfolio population as the primary evolving population and the finite scenario archive as a dynamic risk-evaluation environment. At each archive update, the scenario side performs candidate generation, vulnerability evaluation, and coverage-based selection. The portfolio side runs standard NSGA-II under the current archive. The two components form a closed loop through the Pareto set and immediate reevaluation of the risk objective.

In the current implementation, PCSC is run independently on each decision day D_m , and the scenario archive is reconstructed rather than inherited across months. The outer-level procedure over the full investment horizon is as follows:

- After the close of D_m , run PCSC using $\mathbf{R}_t$ and $\mathbf{w}_t^- = \mathbf{w}_m^{\text{dec}, -}$ available by that date to obtain and freeze $\mathbf{w}_m^*$;
- Continue holding the previous portfolio until the close of E_m , compute the pre-execution weights according to Eq. (8), rebalance to $\mathbf{w}_m^*$, and deduct costs according to Eq. (9);
- The new portfolio begins accruing returns on S_m and remains invested until the close of the next execution day E_{m+1} . After the close of the preceding trading day D_{m+1} , generate the next target portfolio using information realized by that time;
- After including the final return for the current period at the close of E_{m+1} , execute the next rebalancing and repeat the procedure until the end of the prespecified and frozen test period.

Thus, information from the previous period enters the current optimization only through the drifted weights on the decision day, the return matrix available by that date, and parameters frozen during validation. The execution-day return affects only the realized pretrade weights and out-of-sample accounting and is not fed back into the completed search.

3.1 Overall Execution Order and Initialization

Within a rebalancing period, PCSC performs the following five stages in sequence:

- Initialize and repair the portfolio population $\mathcal{P}^0$;
- Randomly generate the first batch of candidate scenarios and construct the initial scenario archive $\mathcal{A}^0$ using the entire $\mathcal{P}^0$;
- Compute both portfolio objectives using $\mathcal{A}^0$ and establish the first valid non-dominated ranking;
- Evolve the portfolio population and update the scenario archive every K generations according to the current Pareto set;

- Output the final Pareto set and select the portfolio for execution in the next holding period according to Eq. (31).

Initialization requires separate treatment because PCSC contains a circular dependency: identifying the Pareto set requires the archive-based risk objective, whereas constructing a Pareto-set-aware archive requires the Pareto set. To break this cycle, the first archive update uses the entire feasible initial population as the feedback set rather than the not-yet-available Pareto set. After the first formal evaluation, all subsequent archive updates use the current first non-dominated front.

The i th component of each raw initial weight vector $\mathbf{z}_j$ is sampled independently and uniformly from $[0, u_{t,i}]$. The constraint-repair procedure in Section 3.5 then produces $\mathbf{w}_j \in \mathcal{W}_t$. The initialization procedure is given in Algorithm 1.

Algorithm 1 PCSC Initialization

Require: $\mathbf{R}_t$, $\mathbf{w}_t^-$, population size N , number of candidates Q , archive capacity M , and portfolio and scenario bounds

Ensure: $\mathcal{P}^0$, $\mathcal{A}^0$, and the initial objective matrix

- 1: $\mathcal{P}^0 \leftarrow \emptyset$
 - 2: **for** $j = 1, \dots, N$ **do**
 - 3: $z_{j,i} \sim \text{Unif}([0, u_{t,i}])$, $i = 1, \dots, n$
 - 4: $\mathbf{w}_j \leftarrow \text{WeightRepair}(\mathbf{z}_j, \mathbf{w}_t^-, (u_{t,i})_{i=1}^n, \tau_{\max})$
 - 5: $\mathcal{P}^0 \leftarrow \mathcal{P}^0 \cup \{\mathbf{w}_j\}$
 - 6: **end for**
 - 7: $\mathcal{U}^0 \leftarrow \mathcal{P}^0$ ▷ The Pareto set is not yet available
 - 8: $\mathcal{A}^0 \leftarrow \text{ArchiveUpdate}(\mathbf{R}_t, \mathcal{U}^0, \emptyset, Q, M)$
 - 9: Evaluate the entire $\mathcal{P}^0$ using $\mathcal{A}^0$ and Eq. (17)
-

3.2 Scenario Candidate Generation and Repair

Let Q be the number of temporary candidates and M the capacity of the formal archive, with $Q \geq M$. Candidate generation differs between the initial and subsequent updates.

3.2.1 Initial Generation

When the previous archive is empty, all Q candidates are generated independently at random:

$$\begin{aligned} b &\sim \text{Unif}_{\mathbb{Z}}\{0, \dots, T - H\} \\ a &\sim \text{Unif}[a_{\min}, a_{\max}] \\ \rho &\sim \text{Unif}[\rho_{\min}, \rho_{\max}] \end{aligned} \tag{32}$$

No parent scenario is available at this stage; therefore, p_{imm} and p_{reset} are not used.

3.2.2 Subsequent Generation

When the previous archive $\mathcal{A}^{g-1}$ is nonempty, repaired copies of its M scenarios are first inserted into the candidate sequence. They are only given another opportunity to participate in selection and are not guaranteed admission to the new archive. The remaining $Q - M$ positions are then filled sequentially. For each empty position, p_{imm} controls the balance between global exploration and neighborhood search around the archive through the following two branches:

- Random immigrant: with probability p_{imm} , generate a random immigrant independent of the previous archive according to Eq. (32);
- Archive mutation: with probability $1 - p_{\text{imm}}$, select a parent scenario uniformly from the previous archive and apply mixed discrete-continuous mutation.

For the archive-mutation branch, let the parent scenario be $s_k = (b_k, a_k, \rho_k)$. With probability p_{reset} , the block starting index is resampled from all valid starting indices; otherwise, it is shifted locally around the parent. The

two continuous parameters receive Gaussian perturbations in both cases:

$$\begin{aligned}
b'_k &= \begin{cases} \tilde{b}_k, & \text{with probability } p_{\text{reset}} \\ b_k + \delta_b, & \text{with probability } 1 - p_{\text{reset}} \end{cases} \\
\tilde{b}_k &\sim \text{Unif}_{\mathbb{Z}}\{0, \dots, T - H\} \\
\delta_b &\sim \text{Unif}_{\mathbb{Z}}\{-R_b, \dots, R_b\} \\
a'_k &= a_k + \xi_a, \quad \xi_a \sim \mathcal{N}(0, [\eta_s(a_{\max} - a_{\min})]^2) \\
\rho'_k &= \rho_k + \xi_\rho, \quad \xi_\rho \sim \mathcal{N}(0, [\eta_s(\rho_{\max} - \rho_{\min})]^2)
\end{aligned} \tag{33}$$

where R_b is the local shift radius for the block starting index and η_s is the continuous mutation scale relative to the parameter-range width. For an unfilled candidate position, the overall probabilities of the three operations are

$$\begin{aligned}
\Pr(\text{random immigrant}) &= p_{\text{imm}} \\
\Pr(\text{archive mutation with starting-index reset}) &= (1 - p_{\text{imm}})p_{\text{reset}} \\
\Pr(\text{archive mutation with local starting-index shift}) &= (1 - p_{\text{imm}})(1 - p_{\text{reset}})
\end{aligned} \tag{34}$$

All randomly generated or mutated scenarios are repaired using Eq. 35. This operation ensures that the block starting index corresponds to a complete historical window and that the continuous parameters remain within their prespecified bounds. The candidate-generation procedure is given in Algorithm 2.

$$\mathcal{R}_s(b, a, \rho) = (\text{clip}(\text{round}(b), 0, T - H), \text{clip}(a, a_{\min}, a_{\max}), \text{clip}(\rho, \rho_{\min}, \rho_{\max})) \tag{35}$$

Algorithm 2 Stress Scenario Candidate Generation

Require: Previous archive $\mathcal{A}^{g-1}$, number of candidates Q , return-window length T , scenario parameter bounds, p_{imm} , p_{reset} , R_b , and η_s

Ensure: Ordered candidate sequence $\mathcal{S}^g = (s_1, \dots, s_Q)$

- 1: $\mathcal{S}^g \leftarrow$ repaired copies of the scenarios in the previous archive
 - 2: **while** $|\mathcal{S}^g| < Q$ **do**
 - 3: **if** $\mathcal{A}^{g-1} = \emptyset$ **then**
 - 4: Randomly generate s according to Eq. (32)
 - 5: **else if** A random immigrant is triggered with probability p_{imm} **then**
 - 6: Randomly generate s according to Eq. (32)
 - 7: **else**
 - 8: Select parent scenario s_{par} uniformly from $\mathcal{A}^{g-1}$
 - 9: Generate mutated scenario s according to Eq. (33)
 - 10: **end if**
 - 11: Append $\mathcal{R}_s(s)$ to $\mathcal{S}^g$
 - 12: **end while**
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3.3 Vulnerability Evaluation and Coverage-Based Archive Selection

The inputs to a scenario archive update are the portfolio feedback set $\mathcal{U}^g = \{\mathbf{w}_j\}_{j=1}^J$, the previous archive, and the newly generated candidate sequence. At initialization, $\mathcal{U}^0 = \mathcal{P}^0$; thereafter, $\mathcal{U}^g$ is the first non-dominated front of the current population immediately before the archive update.

We first compute two loss matrices: the candidate stress-loss matrix D_{jk} and the unperturbed historical-baseline loss matrix E_{jb} :

$$D_{jk} = L(\mathbf{w}_j, s_k), \quad D \in \mathbb{R}^{J \times Q} \tag{36}$$

$$E_{jb} = L(\mathbf{w}_j, (b, 1, 0)), \quad E \in \mathbb{R}^{J \times B} \tag{37}$$

The median and MAD are computed separately for each row of E , after which Eq. (24) yields the vulnerability matrix $V = (v_{jk}) \in \mathbb{R}_+^{J \times Q}$.

The archive is constructed greedily from the empty set. Let c_j denote the maximum vulnerability coverage attained for portfolio j by the currently selected scenarios, initialized as $c_j = 0$. For an unselected candidate s_k , its marginal gain can be written directly as

$$\Delta_k = \frac{1}{J} \sum_{j=1}^J [\max\{c_j, v_{jk}\} - c_j] \tag{38}$$

After selecting the candidate with the largest Δ_k , update $c_j \leftarrow \max\{c_j, v_{jk}\}$ and recompute the gains of the remaining candidates. Because $c_j = 0$ in the first round, the scenario with the largest mean normalized vulnerability is selected. From the second round onward, portfolios already covered by the archive do not continue to provide the same gain to scenarios that repeatedly attack them. The mechanism therefore allocates the limited archive capacity to complementary weaknesses across different Pareto portfolios, rather than simply retaining several similar scenarios with the largest average losses.

Algorithm 3 presents the complete archive update, comprising candidate generation, portfolio-wise normalization, and greedy selection. If multiple candidates have the same marginal gain, the candidate with the smallest index in the original sequence is selected.

Algorithm 3 Scenario Archive Update Based on Pareto Vulnerability Coverage

Require: $\mathbf{R}_t$, feedback set $\mathcal{U}^g$, previous archive $\mathcal{A}^{g-1}$, number of candidates Q , and archive capacity M

Ensure: New archive $\mathcal{A}^g$, coverage trajectory, and indices of archived candidates

```

1:  $\mathcal{S}^g \leftarrow \text{CandidateGeneration}(\mathcal{A}^{g-1}, Q)$ 
2: Construct all unperturbed historical blocks  $\mathcal{B}_t$ 
3: for each  $\mathbf{w}_j \in \mathcal{U}^g$  do
4:   for each  $s_k \in \mathcal{S}^g$  do
5:      $D_{jk} \leftarrow L(\mathbf{w}_j, s_k)$ 
6:   end for
7:   for each  $b \in \mathcal{I}_t$  do
8:      $E_{jb} \leftarrow L(\mathbf{w}_j, (b, 1, 0))$ 
9:   end for
10:  Compute  $\mu_j$  and  $\sigma_j$  according to Eq. (23)
11:  Compute  $v_{jk}$  ( $k = 1, \dots, Q$ ) according to Eq. (24)
12: end for
13:  $\mathcal{A}^g \leftarrow \emptyset$ ,  $c_j \leftarrow 0$  ( $j = 1, \dots, J$ ),  $\mathcal{I} \leftarrow \{1, \dots, Q\}$ 
14: for  $\ell = 1, \dots, M$  do
15:   Compute  $\Delta_k$  for all  $k \in \mathcal{I}$  according to Eq. (38)
16:    $k^* \leftarrow \min(\arg \max_{k \in \mathcal{I}} \Delta_k)$ 
17:   Append  $s_{k^*}$  to  $\mathcal{A}^g$ 
18:    $c_j \leftarrow \max\{c_j, v_{jk^*}\}$ ,  $j = 1, \dots, J$ 
19:   Save the current marginal gain and  $C(\mathcal{A}^g; \mathcal{U}^g)$ 
20:    $\mathcal{I} \leftarrow \mathcal{I} \setminus \{k^*\}$ 
21: end for

```

3.4 Portfolio Evolution with Bidirectional Feedback

After initialization, the portfolio population runs NSGA-II under the current scenario archive. Each generation performs non-dominated sorting, crowding-distance calculation, binary tournament selection, Simulated Binary Crossover (SBX), polynomial mutation, and environmental selection. The polynomial mutation probability for each weight variable is $1/n$; the crossover probability, distribution indices, and other numerical parameters are determined using only the validation period. The scenario archive is updated every K generations. An archive-update generation contains feedback in both directions:

- Portfolio-to-scenario feedback: first identify the current first non-dominated front under the previous archive $\mathcal{A}^{g-1}$ and use this front as $\mathcal{U}^g$. Candidate-scenario vulnerabilities and coverage gains are therefore determined by the currently competitive portfolios.
- Scenario-to-portfolio feedback: after obtaining the new archive $\mathcal{A}^g$, immediately use it to recompute the second objective for the entire current population, followed by non-dominated sorting and crowding-distance calculation. Parent tournament selection is performed only after this immediate reevaluation. Otherwise, parent selection would retain the ranks and crowding distances produced by the previous archive, preventing the new scenarios from changing the evolutionary pressure in the current generation.

In generations without an archive update, set $\mathcal{A}^g = \mathcal{A}^{g-1}$. Regardless of whether the archive is updated, offspring in the current generation are evaluated using $\mathcal{A}^g$, and environmental selection from the combined parent-offspring population is based on the same archive. The complete procedure is given in Algorithm 4.

The coevolution in this procedure is not a simple sequential combination of two independent optimizers. The current Pareto set changes the portfolios against which candidate scenarios are evaluated and the order in which they

Algorithm 4 Pareto-Coverage Stress Coevolution

Require: $\mathbf{R}_t, \mathbf{w}_t^-, (u_{t,i})_{i=1}^n, N, G, K, Q, M$, and portfolio and scenario parameters

Ensure: Final Pareto set, portfolio selected for execution, and final scenario archive

- 1: Obtain $\mathcal{P}^0, \mathcal{A}^0$, and objective values using Algorithm 1
 - 2: **for** $g = 1, \dots, G$ **do**
 - 3: Perform non-dominated sorting and compute crowding distances for $\mathcal{P}^{g-1}$ under $\mathcal{A}^{g-1}$
 - 4: $\mathcal{A}^g \leftarrow \mathcal{A}^{g-1}$
 - 5: **if** $g \bmod K = 0$ **then**
 - 6: $\mathcal{U}^g \leftarrow \text{FirstFront}(\mathcal{P}^{g-1}, \mathcal{A}^{g-1})$
 - 7: $\mathcal{A}^g \leftarrow \text{ArchiveUpdate}(\mathbf{R}_t, \mathcal{U}^g, \mathcal{A}^{g-1}, Q, M)$
 - 8: Reevaluate the entire $\mathcal{P}^{g-1}$ using $\mathcal{A}^g$
 - 9: Recompute non-domination ranks and crowding distances
 - 10: **end if**
 - 11: Select parents from $\mathcal{P}^{g-1}$ by binary tournament
 - 12: Generate N raw offspring using SBX and polynomial mutation
 - 13: Apply WeightRepair to each raw offspring to obtain $\mathcal{O}^g$
 - 14: Evaluate $\mathcal{O}^g$ using $\mathcal{A}^g$
 - 15: Combine $\mathcal{P}^{g-1}$ and $\mathcal{O}^g$
 - 16: Select N individuals by non-domination rank and crowding distance to obtain $\mathcal{P}^g$
 - 17: **end for**
 - 18: $\mathcal{P}_G^* \leftarrow \text{FirstFront}(\mathcal{P}^G, \mathcal{A}^G)$
 - 19: Select the portfolio for execution from $\mathcal{P}_G^*$ according to Eq. (31)
-

enter the archive; the updated archive, in turn, changes the risk objectives, dominance relations, and reproductive opportunities of all portfolios. Removing feedback in either direction changes the core search process.

3.5 Portfolio Constraint Repair, Output, and Reproducibility

Capped-simplex projection. Let $\mathbf{z}$ be the raw weight vector produced by the genetic operators. We first compute its Euclidean projection onto the set defined by the long-only constraint, budget constraint, and time-varying weight caps:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \frac{1}{2} \|\mathbf{w} - \mathbf{z}\|_2^2 \quad \text{s.t.} \quad \mathbf{1}^\top \mathbf{w} = 1, \quad 0 \leq w_i \leq u_{t,i} \quad (39)$$

The projection satisfies

$$\hat{w}_i = \text{clip}(z_i - \lambda, 0, u_{t,i}) \quad (40)$$

where λ is obtained by bisection such that $\sum_i \hat{w}_i = 1$. If $\sum_i u_{t,i} < 1$, the weight caps are incompatible with the budget constraint, and the algorithm reports infeasibility directly.

Feasible anchor under drifted weights. Price changes during the holding period or changes in asset availability may cause some components of $\mathbf{w}_t^-$ to exceed their corresponding caps $u_{t,i}$. Direct interpolation between $\mathbf{w}_t^-$ and $\hat{\mathbf{w}}$ does not guarantee that the entire line segment satisfies the weight caps. We therefore first construct a feasible anchor with the smallest attainable turnover distance from $\mathbf{w}_t^-$. The components are first clipped as follows:

$$q_i^{(0)} = \min\{w_{t,i}^-, u_{t,i}\}, \quad D_0 = 1 - \sum_i q_i^{(0)} \quad (41)$$

D_0 is the budget that must be reallocated after releasing weight from assets above their caps. Let

$$p_i = \max\{\hat{w}_i - q_i^{(0)}, 0\}, \quad h_i^{(0)} = u_{t,i} - q_i^{(0)} \quad (42)$$

If $\sum_i p_i > 0$, allocate the budget first along the positive components preferred by the candidate projection:

$$q_i^{(1)} = q_i^{(0)} + \min \left\{ \frac{D_0 p_i}{\sum_l p_l}, h_i^{(0)} \right\} \quad (43)$$

Otherwise, set $\mathbf{q}^{(1)} = \mathbf{q}^{(0)}$. If a residual budget $D_1 = 1 - \sum_i q_i^{(1)} > 0$ remains, allocate it in proportion to the remaining capacity:

$$w_i^{\text{anc}} = q_i^{(1)} + D_1 \frac{u_{t,i} - q_i^{(1)}}{\sum_l (u_{t,l} - q_l^{(1)})} \quad (44)$$

The resulting $\mathbf{w}^{\text{anc}}$ satisfies nonnegativity, the budget constraint, and all per-asset caps. If $\tau(\mathbf{w}^{\text{anc}}, \mathbf{w}_t^-) > \tau_{\max}$, correcting the cap violations caused by drift alone already exceeds the turnover budget, and the current constraint set is infeasible. The implementation reports an error rather than silently relaxing $\tau_{\max}$.

Turnover repair along a line segment. If $\hat{\mathbf{w}}$ satisfies the turnover cap, it is returned directly. Otherwise, the largest feasible step is found along the line segment from the feasible anchor to the candidate projection:

$$\alpha^* = \max_{0 \leq \alpha \leq 1} \alpha, \quad \text{s.t.} \quad \tau(\mathbf{w}^{\text{anc}} + \alpha(\hat{\mathbf{w}} - \mathbf{w}^{\text{anc}}), \mathbf{w}_t^-) \leq \tau_{\max}, \quad \mathbf{w}^{\text{rep}} = \mathbf{w}^{\text{anc}} + \alpha^*(\hat{\mathbf{w}} - \mathbf{w}^{\text{anc}}) \quad (45)$$

The capped simplex is convex; therefore, every point on the line segment satisfies nonnegativity, the budget constraint, and the weight caps. α^* is obtained by bisection. The complete repair procedure is given in Algorithm 5.

Algorithm 5 Portfolio Weight and Turnover Constraint Repair

Require: Raw vector $\mathbf{z}$, weight-cap vector $(u_{t,i})_{i=1}^n$, and optional $\mathbf{w}_t^-$ and $\tau_{\max}$

Ensure: Repaired weights $\mathbf{w}^{\text{rep}}$

- 1: **if** $\sum_i u_{t,i} < 1$ **then**
 - 2: Report infeasibility of the budget and weight caps and terminate
 - 3: **end if**
 - 4: Compute $\hat{\mathbf{w}}$ according to Eq. (39)
 - 5: **if** $\mathbf{w}_t^-$ does not exist **then**
 - 6: **return** $\hat{\mathbf{w}}$
 - 7: **end if**
 - 8: Construct $\mathbf{w}^{\text{anc}}$ according to Eqs. (41)–(44)
 - 9: **if** $\tau(\mathbf{w}^{\text{anc}}, \mathbf{w}_t^-) > \tau_{\max}$ **then**
 - 10: Report infeasibility of the turnover constraint and terminate
 - 11: **end if**
 - 12: **if** $\tau(\hat{\mathbf{w}}, \mathbf{w}_t^-) \leq \tau_{\max}$ **then**
 - 13: **return** $\hat{\mathbf{w}}$
 - 14: **end if**
 - 15: Solve for α^* in Eq. (45) by bisection
 - 16: **return** $\mathbf{w}^{\text{anc}} + \alpha^*(\hat{\mathbf{w}} - \mathbf{w}^{\text{anc}})$
-

Output and interperiod execution. After G generations, PCSC retains the complete final Pareto set and selects a unique target portfolio $\mathbf{w}_m^*$ according to Eq. (31). This portfolio is executed and evaluated following the timing protocol in Section 2.1. The next PCSC run starts independently at D_{m+1} using the information then available, while the current portfolio remains invested until the close of E_{m+1} . Neither the selection rule nor the algorithm parameters are modified according to out-of-sample performance during the test period.

Computational complexity and reproducibility records. Let J be the number of feedback portfolios and $B = T - H + 1$ the number of historical blocks. During one archive update, the naive time complexity of computing candidate stress losses and historical baseline losses is $O(J(Q + B)Hn)$, while greedy coverage selection requires $O(JQM)$. Evaluating N portfolios under a fixed archive in each generation requires $O(NMHn)$, and standard non-dominated sorting for two objectives requires $O(N^2)$. The implementation preconstructs the unperturbed historical blocks and caches candidate stressed-return paths within each archive update.

The formal experimental runner records the random seed, complete configuration, final weights, objective values, Pareto set size in each generation, candidate indices at archive updates, marginal coverage gains, and coverage trajectories. All tied selections use a stable index order, ensuring that identical data, configurations, and random seeds produce a unique reproducible experimental trajectory.

4 Complete Experimental Setup

4.1 Temporal Split and Walk-Forward Backtest

The data split follows strict chronological order: 2016–2018 is used for data auditing, point-in-time stock-universe construction, and code development; 2019 is used only for execution-correctness checks and parameter freezing; and 2020–2025 is used for a one-time Out-of-Sample Test (OOS), with no parameter selection. The validation and test

periods start independently from cash, preventing validation-period holdings or random states from carrying over into the test period. One-way turnover at initial investment is recorded as 1. Thereafter, each method computes the pretrade weights on the execution day, realized turnover, and transaction costs along its own holding path.

The backtest follows the monthly protocol in Section 2.1. Execution occurs at the close of the first trading day of each month, with the close of the preceding trading day serving as the decision time. The estimation window contains the most recent $T = 252$ daily returns available by the decision day. Target weights are frozen before execution, begin accruing returns on the trading day following the execution day, and remain invested until the close of the next month’s execution day. The 2020–2025 period contains 72 rebalancing cycles. The main experiment uses a one-way proportional transaction cost of 10 bps. The ex ante cost in the optimization objective and the realized cost in the backtest use the same rate but are based on the drifted weights on the decision and execution days, respectively; see Eqs. (5) and (9) for the distinction.

4.2 Compared Methods

Table 2 summarizes the six methods in the main experiment. All randomized multi-objective methods use the same portfolio encoding, constraint repair, NSGA-II budget, and normalized-ideal-point rule for selecting the portfolio for execution. The methods differ only in their risk objectives or scenario archive mechanisms.

Table 2: Methods in the main experiment and their risk-evaluation schemes

Method	Risk evaluation and scenario mechanism	Runs
Equal Weight (EW)	Equal weights over currently available assets, using the common weight and turnover repair procedure	1
Minimum Variance (MinVar)	252-day sample covariance; the minimum-variance portfolio is solved after adding a 10^{-6} ridge scaled by the mean variance	1
NSGA-II Mean-CVaR	Maximizes expected net return while minimizing the 95% CVaR of historical daily losses	20
Fixed Worst-Block NSGA-II	Selects M fixed historical blocks by equally weighted market path loss; the archive is not updated within a monthly optimization	20
AvgWorst Coevolution	Uses the PCSC scenario representation and update frequency but selects scenarios by mean raw path loss over the entire population	20
PCSC	Uses Pareto set feedback, Portfolio-wise Normalized Vulnerability, marginal maximum coverage, and immediate risk reevaluation after archive updates	20

EW and MinVar are deterministic and are not replicated as pseudorandom runs. Fixed Worst-Block selects the M historical blocks with the largest equally weighted portfolio path losses from all valid blocks, allowing windows to overlap; its scenarios are independent of the current portfolio population. AvgWorst Coevolution is a matched baseline constructed using the common representation in this work. It distinguishes “attacking the entire population by mean raw loss” from the Pareto Vulnerability Coverage mechanism of PCSC and is not presented as an exact reproduction of a published coevolutionary algorithm.

4.3 Ablations and Transaction-Cost Sensitivity

Full PCSC and the two ablations use the same random seeds, number of candidates, archive capacity, and optimization budget:

- **w/o portfolio-to-scenario feedback** (hereafter **w/o Feedback**): the initial archive is still constructed using the coverage criterion but remains fixed thereafter. Thus, only the persistent feedback from the current Pareto set to the scenario archive is removed; the fixed archive remains in use for portfolio risk evaluation.
- **w/o marginal coverage** (hereafter **w/o Marginal Coverage**): Pareto set feedback, portfolio-wise median/MAD normalization, and immediate reevaluation after archive updates are retained, but greedy marginal maximum coverage is replaced by ranking scenarios according to mean normalized vulnerability over the Pareto portfolios.

The second ablation tests marginal set coverage rather than normalization itself. The claim concerning normalization is limited to reducing the influence of differences in absolute risk scales across portfolios on scenario evaluation.

Transaction-cost sensitivity is evaluated at $c \in \{0, 10, 30\}$ bps. The main 10-bps results are reused directly, and only PCSC and its closest baseline, AvgWorst Coevolution, are additionally run at 0 and 30 bps. At each cost level, the optimization objective and out-of-sample cost deduction use the same c , with all other settings held fixed.

4.4 Algorithm Parameters and Validation-Period Freezing

Table 3 summarizes the prespecified parameters. Using five predetermined seeds in 2019, we first verify that portfolio and scenario constraints have no violations, objective values are finite, the Pareto set is nonempty, the archive capacity is correct, and runs with identical configurations and seeds are reproducible. If the scales reported in the table pass these checks, they are frozen directly without a large-scale grid search. Only in the event of a clear algorithmic failure may one relevant parameter be adjusted using 2019 data alone, with the reason for the change recorded. No parameter may be modified according to the 2020–2025 results.

Table 3: Prespecified PCSC and common experimental parameters

Category	Parameter	Value
Data and trading	Number of stocks n ; estimation window T ; rebalancing	50; 252 days; monthly
Investment constraints	u ; $\tau_{\max}$; main cost c	0.10; 0.30; 10 bps
Stress scenarios	H ; a_k ; ρ_k	20; [1.0, 1.5]; [0, 0.5]
Scenario archive	M ; Q ; update interval K	12; 24; 5 generations
Scenario mutation	p_{imm} ; p_{reset} ; local block-shift radius; continuous perturbation scale	0.25; 0.20; $H = 20$ days; 0.10 of the parameter-range width
Numerical stability	Vulnerability-normalization stabilizer ϵ	10^{-8}
NSGA-II	Population N ; generations G	80; 50
Genetic operators	SBX; polynomial mutation	$p_c = 0.9, \eta_c = 15$; $p_m = 1/n, \eta_m = 20$
Execution solution	Pareto solution selection	Closest to the normalized ideal point
Statistical replications	Number of randomized runs	20 paired run seeds

All constraint, finiteness, nonempty-Pareto-Set, archive-capacity, and reproducibility checks pass for the five predetermined validation seeds. After ignoring run-identity and state-gating fields, the validation and test records have identical scientific configurations. The parameters in Table 3 were therefore not modified using test-period results and were frozen before the 2020–2025 test gate was opened.

4.5 Evaluation Metrics and Mechanism Diagnostics

Let $x_1, \dots, x_D$ denote the daily OOS net simple returns and $V_d = \prod_{q=1}^d (1 + x_q)$ the portfolio value. Annualized net return, annualized volatility, the Sharpe ratio, and the Sortino ratio are defined as follows:

$$\begin{aligned}
 R_{\text{ann}} &= V_D^{252/D} - 1, & \sigma_{\text{ann}} &= \sqrt{252} \text{sd}(x), \\
 \text{Sharpe} &= \sqrt{252} \frac{\bar{x}}{\text{sd}(x)}, & \text{Sortino} &= \sqrt{252} \frac{\bar{x}}{\sqrt{D^{-1} \sum_{d=1}^D \min(x_d, 0)^2}}
 \end{aligned} \tag{46}$$

The risk-free rate and the minimum acceptable return for the Sortino ratio are both set to zero. Maximum Drawdown and Worst 20-Day Cumulative Return (W20R) are defined as

$$\begin{aligned}
 \text{MDD} &= \max_{1 \leq d \leq D} \left(1 - \frac{V_d}{\max_{0 \leq q \leq d} V_q} \right), & V_0 &= 1 \\
 R_{20}^{\text{worst}} &= \min_{20 \leq d \leq D} \left[\prod_{q=d-19}^d (1 + x_q) - 1 \right]
 \end{aligned} \tag{47}$$

The historical 95% CVaR is the negative mean of the worst 5% of OOS daily returns and is reported as a positive loss. Average One-Way Turnover (AOWT) is the arithmetic mean of realized monthly one-way turnover. The main text reports annualized net return, Sharpe ratio, MDD, CVaR, W20R, and AOWT; the supplementary material additionally reports annualized volatility and the Sortino ratio.

To compare archive quality among the three PCSC variants over the same portfolio set, for each period m and run seed r , we combine the final Pareto sets from the full method and the two ablations as follows:

$$\mathcal{U}_{m,r}^{\text{ref}} = \mathcal{P}_{m,r}^{*,\text{PCSC}} \cup \mathcal{P}_{m,r}^{*,\text{NF}} \cup \mathcal{P}_{m,r}^{*,\text{NC}} \quad (48)$$

where NF and NC denote the variants without persistent feedback and without marginal set coverage, respectively. The coverage and threshold coverage rate of the final archive $\mathcal{A}$ from any variant over this Common Reference Portfolio Set are

$$\begin{aligned} C_{\text{ref}}(\mathcal{A}) &= \frac{1}{|\mathcal{U}_{m,r}^{\text{ref}}|} \sum_{\mathbf{w} \in \mathcal{U}_{m,r}^{\text{ref}}} \max_{s \in \mathcal{A}} v_+(\mathbf{w}, s) \\ \text{PCR}_1(\mathcal{A}) &= \frac{1}{|\mathcal{U}_{m,r}^{\text{ref}}|} \sum_{\mathbf{w} \in \mathcal{U}_{m,r}^{\text{ref}}} \mathbb{I} \left[\max_{s \in \mathcal{A}} v_+(\mathbf{w}, s) \geq 1 \right] \end{aligned} \quad (49)$$

Archive Utilization is defined as the number of archive members that are the worst scenario for at least one reference portfolio, divided by M . In the event of a tie, the archive index determines a unique worst scenario. Each run is first averaged over the 72 test periods and then summarized across the 20 runs. C_{ref} reflects both stress severity and coverage and must therefore be interpreted jointly with the economic metrics and ablation results; it cannot alone serve as evidence of improved investment performance.

4.6 Randomized Runs and Statistical Tests

Randomized methods use the same 20 run identifiers to form pairs. Within each run, independent optimization seeds are derived from the run seed and monthly-period index. For each metric computed over the complete OOS path, we report the median and Interquartile Range (IQR). PCSC is compared separately with the three randomized main baselines and the two ablations using two-sided paired Wilcoxon signed-rank tests. Holm correction is applied to the five comparisons within each reported metric, with a significance level of 0.05. Paired median differences and Cliff’s delta are also reported. When computing effect sizes, metrics for which lower values are preferred, including annualized volatility, MDD, CVaR, and AOWT, are negated so that a positive effect consistently favors PCSC. EW and MinVar have no algorithmic randomness and are included only for descriptive comparison; a single path is not replicated into 20 pseudo-observations.

These randomized runs measure optimization-algorithm randomness rather than 20 independent markets. We do not additionally perform a moving-block bootstrap; therefore, the statistical inference pertains only to algorithmic randomness on the given market path.

5 Complete Experimental Results and Statistical Tests

Except for EW and MinVar, the following tables report medians [IQR] over 20 paired randomized runs. Returns, volatility, MDD, CVaR, W20R, and AOWT are expressed as percentages, whereas Sharpe and Sortino are dimensionless. All results are obtained from the frozen 2020–2025 OOS path, with a main transaction cost of 10 bps.

5.1 Main Experimental Results

Table 4 presents the main results at 10 bps. PCSC achieves the highest median annualized net return of 10.981% and the highest median Sharpe ratio of 0.629. Its Sortino ratio is 0.889, which is comparable to the 0.890 achieved by NSGA-II Mean–CVaR. Relative to the closest baseline, AvgWorst, PCSC improves net return, Sharpe ratio, and Sortino ratio, while its MDD and CVaR are similar to those of AvgWorst, with neither paired difference being significant. These results indicate that PCSC provides more competitive return and risk-adjusted performance under the main transaction-cost setting.

5.2 Paired Statistical Tests

Table 5 reports the complete paired tests with PCSC as the reference. The direction of Δ follows Section 4.6, such that a positive value always favors PCSC. Relative to AvgWorst, PCSC significantly improves annualized net return, Sharpe ratio, and Sortino ratio by 1.324 percentage points, 0.064, and 0.093, respectively. Relative to Fixed Worst-Block, PCSC significantly improves annualized net return, Sharpe ratio, Sortino ratio, and MDD by 2.977

Table 4: Main experimental results over 2020–2025 (10 bps)

Method	Ann. Net Return	Volatility	Sharpe	Sortino	MDD	CVaR	W20R	AOWT
EW	8.776	18.305	0.551	0.786	26.043	2.638	-13.116	4.215
MinVar	4.441	13.408	0.391	0.572	27.006	1.840	-8.366	10.864
NSGA-II Mean-CVaR	10.122 [0.523]	17.974 [0.133]	0.626 [0.026]	0.890 [0.041]	20.929 [1.593]	2.667 [0.026]	-16.705 [0.716]	17.635 [0.334]
Fixed Worst-Block	7.870 [1.193]	19.291 [0.081]	0.489 [0.059]	0.690 [0.085]	26.525 [1.360]	2.869 [0.020]	-15.470 [0.317]	18.583 [0.301]
AvgWorst	9.549 [1.102]	19.594 [0.097]	0.564 [0.052]	0.798 [0.077]	25.709 [1.591]	2.910 [0.029]	-15.694 [0.877]	19.865 [0.322]
PCSC	10.981 [1.303]	19.610 [0.193]	0.629 [0.057]	0.889 [0.081]	25.389 [2.744]	2.907 [0.058]	-16.236 [0.851]	22.541 [0.444]

percentage points, 0.128, 0.189, and 1.026 percentage points, respectively. PCSC also achieves significant advantages over both ablations in annualized net return, Sharpe ratio, and Sortino ratio. These results indicate that, under the current test path and main transaction-cost setting, the complete mechanism has a detectable incremental effect on return generation and risk-adjusted performance.

Table 5: Complete paired statistical tests for PCSC (10 bps)

Metric	Δ	Cliff's δ	p_H	Significant
<i>PCSC vs. AvgWorst</i>				
Annualized net return	1.324	0.575	0.0110	Yes
Annualized volatility	-0.025	0.025	1.0000	No
Sharpe	0.064	0.580	0.0110	Yes
Sortino	0.093	0.575	0.0127	Yes
MDD	-0.228	-0.045	0.9899	No
CVaR	-0.001	0.035	0.9354	No
W20R	-0.541	-0.325	0.1912	No
AOWT	-2.706	-1.000	9.54×10^{-6}	Yes
<i>PCSC vs. Fixed Worst-Block</i>				
Annualized net return	2.977	0.945	9.54×10^{-6}	Yes
Annualized volatility	-0.309	-0.835	3.81×10^{-5}	Yes
Sharpe	0.128	0.945	9.54×10^{-6}	Yes
Sortino	0.189	0.945	9.54×10^{-6}	Yes
MDD	1.026	0.345	0.0332	Yes
CVaR	-0.039	-0.615	0.0108	Yes
W20R	-0.797	-0.675	0.0010	Yes
AOWT	-3.881	-1.000	9.54×10^{-6}	Yes
<i>PCSC vs. NSGA-II Mean-CVaR</i>				
Annualized net return	0.369	0.385	0.0897	No
Annualized volatility	-1.590	-1.000	9.54×10^{-6}	Yes
Sharpe	-0.020	-0.070	0.3683	No
Sortino	-0.036	-0.125	0.2943	No
MDD	-4.764	-0.990	9.54×10^{-6}	Yes
CVaR	-0.246	-1.000	9.54×10^{-6}	Yes
W20R	0.438	0.365	0.4304	No
AOWT	-4.867	-1.000	9.54×10^{-6}	Yes
<i>PCSC vs. w/o Feedback</i>				
Annualized net return	1.163	0.620	0.0005	Yes
Annualized volatility	0.058	0.065	1.0000	No
Sharpe	0.051	0.640	0.0005	Yes
Sortino	0.076	0.620	0.0005	Yes
MDD	0.716	0.165	0.9899	No
CVaR	0.004	0.115	0.9354	No
W20R	-0.413	-0.210	0.2280	No
AOWT	-2.782	-1.000	9.54×10^{-6}	Yes
<i>PCSC vs. w/o Marginal Coverage</i>				

Continued on the next page

Table 5 (continued)

Metric	Δ	Cliff's δ	p_H	Significant
Annualized net return	0.733	0.500	0.0110	Yes
Annualized volatility	0.060	0.150	0.3420	No
Sharpe	0.035	0.495	0.0110	Yes
Sortino	0.050	0.490	0.0127	Yes
MDD	0.324	0.185	0.9899	No
CVaR	0.004	0.185	0.9354	No
W20R	-0.348	-0.235	0.1600	No
AOWT	-2.705	-1.000	9.54×10^{-6}	Yes

5.3 Ablation and Mechanism Diagnostics

Table 6 shows that PCSC achieves higher annualized net return, Sharpe ratio, and Sortino ratio than both ablations. Paired tests indicate that all three metrics differ significantly from both ablations, showing that persistent feedback and marginal set coverage both contribute to shaping the final return-side search outcome. In particular, the full method improves the Sharpe ratio by 0.051 relative to w/o Feedback and by 0.035 relative to w/o Marginal Coverage, providing direct ablation evidence for the incremental contributions of the two core components.

Table 6: PCSC ablation results (10 bps)

Method	Ann. Net Return	Volatility	Sharpe	Sortino	MDD	CVaR	W20R	AOWT
PCSC	10.981 [1.303]	19.610 [0.193]	0.629 [0.057]	0.889 [0.081]	25.389 [2.744]	2.907 [0.058]	-16.236 [0.851]	22.541 [0.444]
w/o Feedback	9.637 [0.893]	19.626 [0.187]	0.566 [0.039]	0.799 [0.054]	25.914 [1.461]	2.915 [0.023]	-15.872 [0.770]	19.563 [0.334]
w/o Marginal Coverage	9.817 [0.957]	19.671 [0.111]	0.575 [0.045]	0.817 [0.065]	26.127 [1.486]	2.917 [0.031]	-15.834 [1.195]	19.754 [0.593]

Table 7 constructs the common reference set using the monthly Pareto solutions from PCSC, w/o Feedback, and w/o Marginal Coverage. The C_{ref} of PCSC is lower than those of w/o Feedback and w/o Marginal Coverage by 2.114 and 2.095, respectively, with Holm-corrected $p \approx 4 \times 10^{-6}$ for both comparisons. Its Archive Utilization is also lower by approximately 1.736 and 1.620 percentage points, respectively, with corrected $p \approx 2.06 \times 10^{-4}$ for both comparisons. PCR_1 is 100% for all three methods. These terminal diagnostics do not support higher terminal common-reference coverage or Archive Utilization for PCSC and therefore cannot directly explain the differences in economic metrics. The ablation results show only that the marginal coverage criterion changes the search outcome; its specific mechanism requires further investigation through more fine-grained analysis of evolutionary trajectories.

Table 7: Mechanism diagnostics on the Common Reference Portfolio Set (10 bps)

Method	C_{ref}	PCR_1 (%)	AU (%)
PCSC	6.597 [0.212]	100.000 [0.000]	13.542 [0.492]
w/o Feedback	8.659 [0.158]	100.000 [0.000]	15.278 [1.215]
w/o Marginal Coverage	8.692 [0.153]	100.000 [0.000]	14.988 [1.100]

5.4 Transaction-Cost Sensitivity

Table 8 presents the transaction-cost sensitivity results. PCSC has higher median annualized net returns than AvgWorst at 0, 10, and 30 bps, reaching 10.683%, 10.981%, and 9.733%, respectively. Its median Sharpe ratio also maintains an advantage in the same direction at all three cost levels. At the main setting of 10 bps, the advantages in net return, Sharpe ratio, and Sortino ratio are all significant. These results indicate that the return-side advantage of PCSC does not arise solely from a zero-cost assumption and persists after accounting for transaction frictions.

5.5 Overall Results and Scope of Applicability

Across the five sets of results, the main advantages of PCSC are concentrated in return and risk-adjusted performance. At 10 bps, it achieves the highest median annualized net return and Sharpe ratio among the six main methods and significantly outperforms AvgWorst, Fixed Worst-Block, and the two core ablations. The transaction-cost sensitivity analysis further shows that its median net return and Sharpe ratio retain an advantage in the same

Table 8: Transaction-cost sensitivity results

Cost (bps)	Method	Ann. Net Return		Sharpe		MDD		CVaR		W20R		AOWT	
0	AvgWorst	10.181	[1.195]	0.594	[0.053]	25.380	[1.234]	2.914	[0.038]	-15.894	[0.944]	20.246	[0.373]
0	PCSC	10.683	[1.496]	0.613	[0.069]	25.790	[0.804]	2.919	[0.074]	-16.495	[1.001]	22.904	[0.372]
10	AvgWorst	9.549	[1.102]	0.564	[0.052]	25.709	[1.591]	2.910	[0.029]	-15.694	[0.877]	19.865	[0.322]
10	PCSC	10.981	[1.303]	0.629	[0.057]	25.389	[2.744]	2.907	[0.058]	-16.236	[0.851]	22.541	[0.444]
30	AvgWorst	9.568	[0.740]	0.565	[0.035]	25.646	[1.493]	2.889	[0.020]	-15.984	[0.548]	19.190	[1.002]
30	PCSC	9.733	[1.270]	0.573	[0.059]	26.709	[1.645]	2.910	[0.070]	-16.502	[1.044]	21.868	[0.368]

direction over AvgWorst at all three cost levels. The ablation results indicate that the complete mechanism does not merely add computational steps but materially changes the portfolio search and the final solution selected for execution.

PCSC does not outperform all methods across every risk metric. Its volatility, MDD, and CVaR are higher than those of Mean-CVaR, and its turnover is also significantly higher. Moreover, terminal C_{ref} and Archive Utilization cannot explain the observed return advantage. We therefore limit our conclusion to improvements in return and risk-adjusted performance under the main transaction-cost setting and do not claim that PCSC dominates across all risk metrics or market conditions.

6 Reproducibility Environment

The experiments were conducted on Apple arm64 with macOS 26.3.1 and Python 3.12.0. Key software versions were NumPy 2.5.2, pandas 2.3.3, SciPy 1.18.0, Matplotlib 3.11.1, PyYAML 6.0.3, pymoo 0.6.2, and BaoStock 0.9.3. Randomized methods use 20 independent run identifiers corresponding to seeds 20260816–20260835. Identical identifiers are paired across methods, and monthly seeds are derived deterministically from the run seed and period index. Deterministic methods are run only once.

All parameters were frozen before the 2020–2025 test period was opened. Each test path contains 72 rebalancing periods and 1,454 daily return observations. All final metrics were independently recomputed from daily net returns for verification, with a maximum numerical error of 1.29×10^{-14} . No violations were found for the portfolio budget, per-asset weights, ex ante turnover, or stress scenario bounds.